THEME: PEOPLE FIRST, AI SECOND • CHALLENGE: BUILDING TRUST IN COMMUNITIES

CAELUM

Centralized Agent Environment for Lifesaving and Urgent Mobility • Team ORIZEN

1 Project Overview

Project Name: CAELUM **Team Name:** ORIZEN **Challenge Area:** Healthcare Logistics & Emergency Access

Kevin Ongaro — FULL-STACK SE • LEAD

System Architecture & Orchestration: Drives the multi-agent pipeline and the precise handoff between deterministic Python workers and the Llama 3.1 8B orchestrator, ensuring strict Pydantic JSON context execution.

Antony Maina — DATA SCIENTIST

Intelligence Layer: Tunes Holt-Winters exponential smoothing parameters for the Forecasting Worker and optimizes chunking and cosine similarity thresholds for the Nemotron RAG Retriever.

Edwin Ngengi — FULL-STACK SE

Core Infrastructure & Backend: Owns the Supabase PostgreSQL schema, `pgvector` implementation, Python backend services, and hard database state validation logic.

Caren Chemjor — FULL-STACK SE • UI LEAD

Frontend & Real-time State: Drives React 18+ components, Tailwind styling, WebSocket inventory syncing, and the Human-in-the-Loop (HITL) Approval Inbox.

2 Problem Statement

Template Format: [Regional hospital transfusion officers and emergency coordinators] experience [preventable blood stockouts during urgent surgeries while nearby hospitals incinerate expiring units] because [inventory data is trapped in isolated facility silos and transfers rely on frantic, 45-minute phone-call chains without real-time logistics or policy visibility]. This makes it difficult for them to [secure compatible donor blood within the critical golden hour of trauma and obstetric emergencies, leading to preventable maternal and accident mortalities]. We want to explore a solution that helps them [proactively predict shortages, deterministically route units approaching expiration, and approve verified emergency transfers in under 5 seconds with absolute clinical trust and zero hallucination risk].

Project Statement: Blood is a perishable, non-fungible biological resource (packed red blood cells expire in 35–42 days). In regional healthcare networks across Kenya, blood banking faces a tragic paradox: sub-county facilities frequently face acute stockouts during postpartum hemorrhages, while tertiary hospitals 35 km away incinerate viable units expiring without surgical demand. CAELUM eliminates this manual telephony chaos with an autonomous, verified multi-agent platform.

3 Target Users & Community Insights

User Segments

Primary Users: Hospital Blood Bank Technologists & Regional Health Network Dispatchers managing daily inventory.

Secondary Users: Emergency surgeons, obstetric clinicians, and cold-chain medical couriers.

Key User Needs

- Instant, cross-hospital inventory visibility by ABO/Rh group and expiry.
- Cold-chain and shelf-life transit mathematical guarantees.
- Automated hospital transfer policy & MOU verification.
- Preserved clinical authority via 1-click human approval cards.

4 Proposed Solution & Key Features

CAELUM is a Hybrid Multi-Agent System (MAS) combining classical statistics, deterministic transit mathematics, and single-shot AI policy synthesis:

1. Predictive Shortage Forecasting

Supabase stock ledger paired with classical Holt-Winters exponential smoothing to predict depletion up to 72 hours in advance.

2. Deterministic Corridors & RAG

Python logistics worker filters valid transit routes against shelf life; NVIDIA Nemotron 3.5 Embed searches hospital transfer SOPs via `pgvector`.

3. Single-Shot HITL Approval Inbox

Llama 3.1 8B synthesizes options into an actionable clinical recommendation card, verified by a hard Python validator for 1-click dispatch.

5 Role of Technology & AI (People First)

Why this Tech? React 18 + Tailwind for low-latency emergency triage; Supabase (`pgvector`) for unified relational + vector integrity; Python Statsmodels for auditable forecasting.

Role of AI: AI does NOT calculate math, routes, or inventory. It serves strictly as a natural-language policy synthesizer. The human clinical officer retains 100% authority.

6 Trust, Safety & Data Privacy

Source Transparency: Cites exact mathematical variables and retrieved clauses from hospital transfer SOPs.

Data Minimization: Zero patient PII; tracks only anonymous unit lots and hospital GPS coordinates.

Sovereign On-Prem Ready: Containerized via local NVIDIA NIM, fully compliant with the *Kenya Data Protection Act (2019)*.

7 Friday Build Plan & Execution Roadmap

THURSDAY	**Orientation, Discovery & Phase 1 Build:** Met, discussed, and ideated. Finalized MAS architecture, designed Supabase schema with `pgvector`, built baseline deterministic Python math workers, and scaffolded React 18 frontend.
09:00 - 11:30 AM	**Building Phase 2:** Integrate Llama 3.1 8B single-shot orchestrator with strict Pydantic JSON validation; implement Nemotron policy embeddings; connect Supabase WebSockets; build Shortage Alerts and Approval Inbox.
11:30 - 12:45 PM	**Tests & Hard Validation:** Unit test deterministic math workers; run adversarial tests (injecting fake unit IDs to verify Python Hard Validator blocks 100% of hallucinations); test WebSocket latency under multi-client loads.
12:45 - 01:30 PM	**Final Touches & Project Completion:** UI polish (animations, dispatch notifications); pitch rehearsal; repository verification; final proposal and codebase submission ahead of the **1:30 PM deadline**.

8 Potential Limitations & Future Scope

Current Limitations:

Prototype models a regional network of up to 10 facilities focusing on packed RBC and platelets. Travel times use regional velocity matrices rather than live GPS traffic feeds.

Future Roadmap:

Integrate BLE cold-chain temperature sensors inside smart transport boxes; connect to autonomous medical drone APIs (Zipline); federate with Kenya National Blood Transfusion Service (KNBTS).